Ques: Top K frequent elements

arr = {7, 10, 11, 5, 2, 5, 5, 7, 11, 8, 9} K = 4

ans = {5, 11, 7, 10}

7 → 2
10 → 1
11 → 2
5 → 3
2 → 1
8 → 1
9 → 1

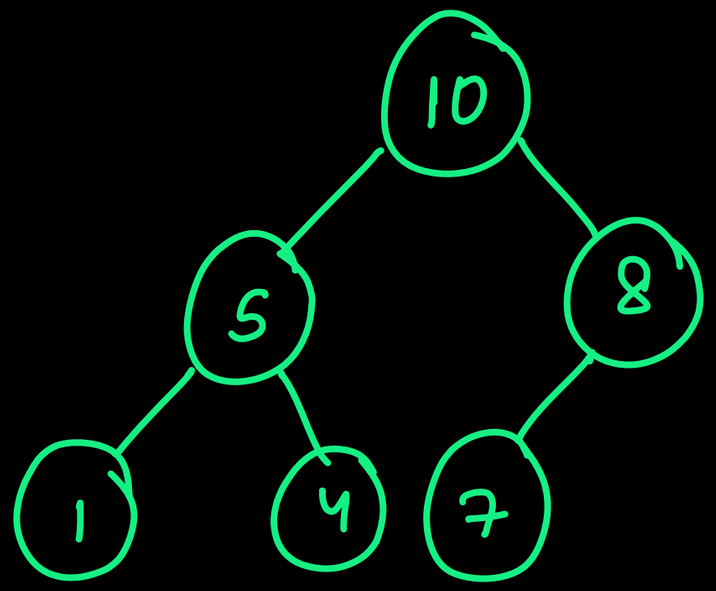
map<ele, freq>

(10, 1)
(11, 2)
(7, 2)
(5, 3)

min heap (K)

(2, 1)
(8, 1)(9, 1)

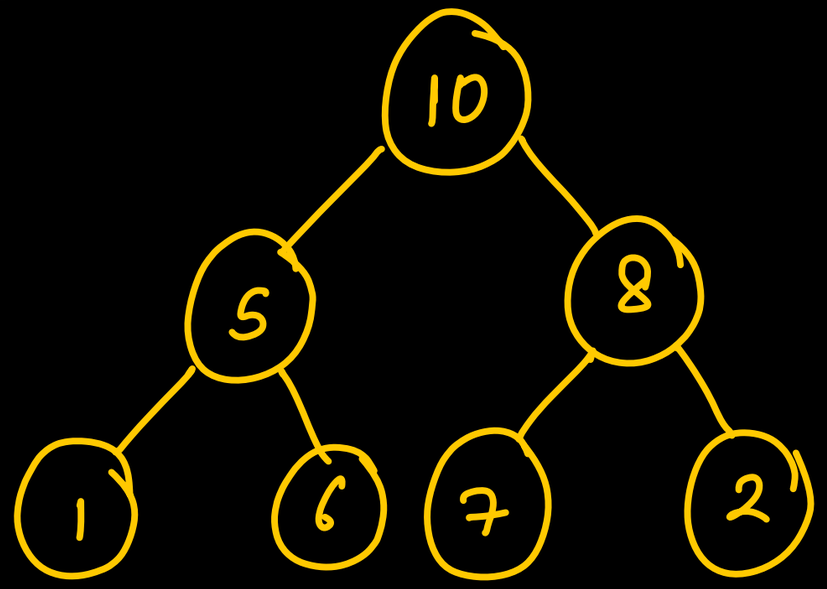
Ques: Is Binary Tree Heap



✓

maxheap property ✓

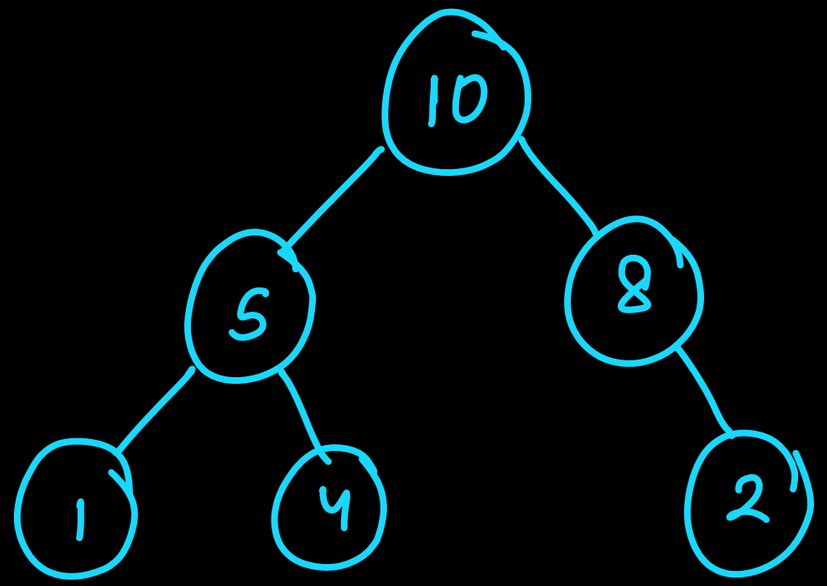
CBT ✓



X

maxheap property X

CBT ✓



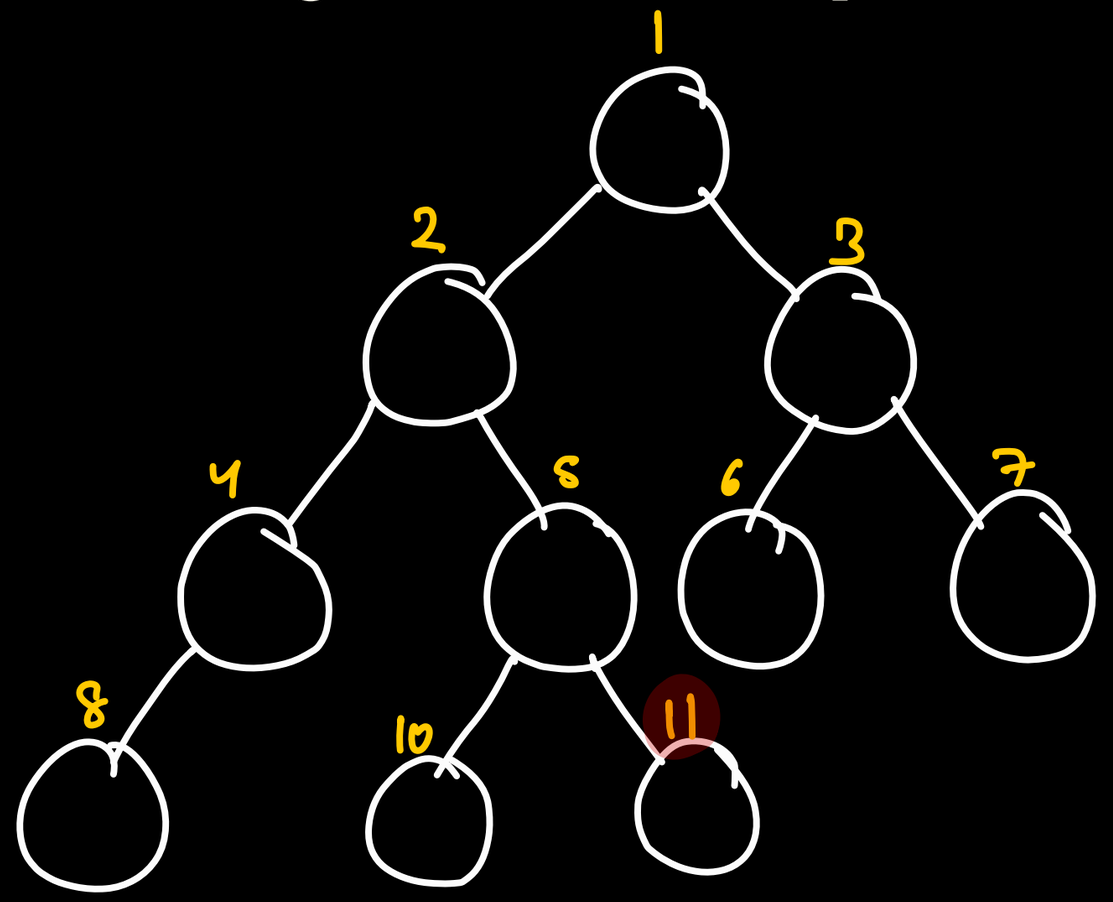
X

maxheap property ✓

CBT X

Ques: Is Binary Tree Heap

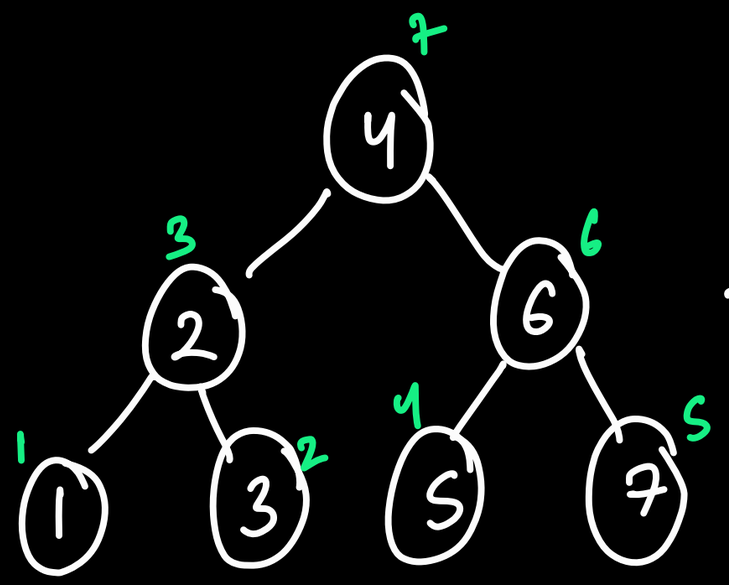
size = 10



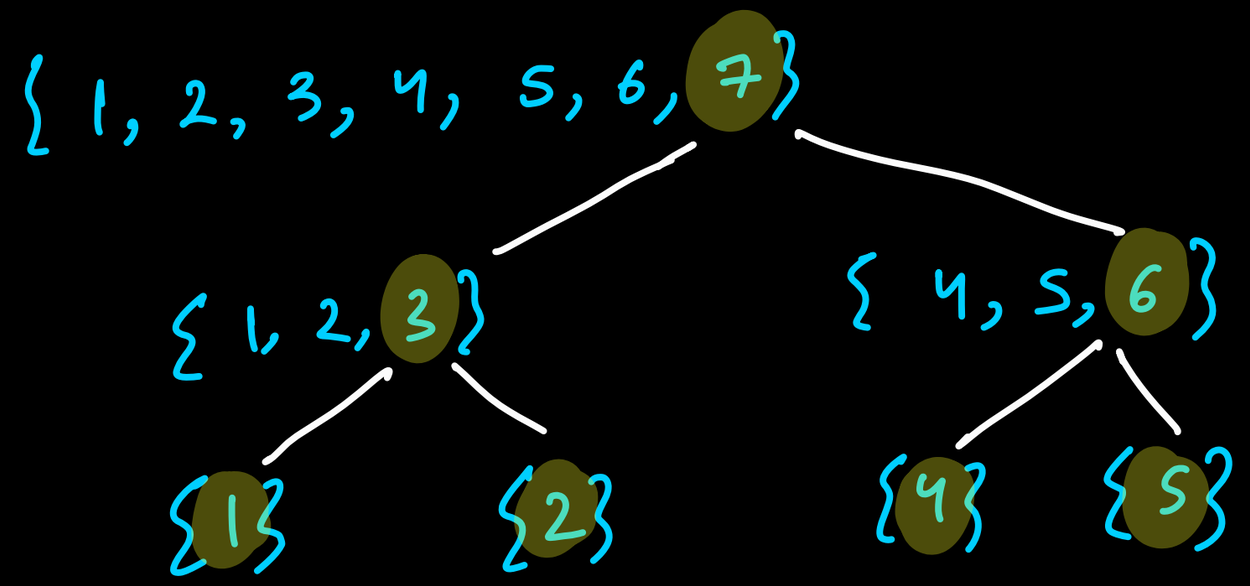
left child = 2i

right child = 2i+1

Ques: BST to Max Heap



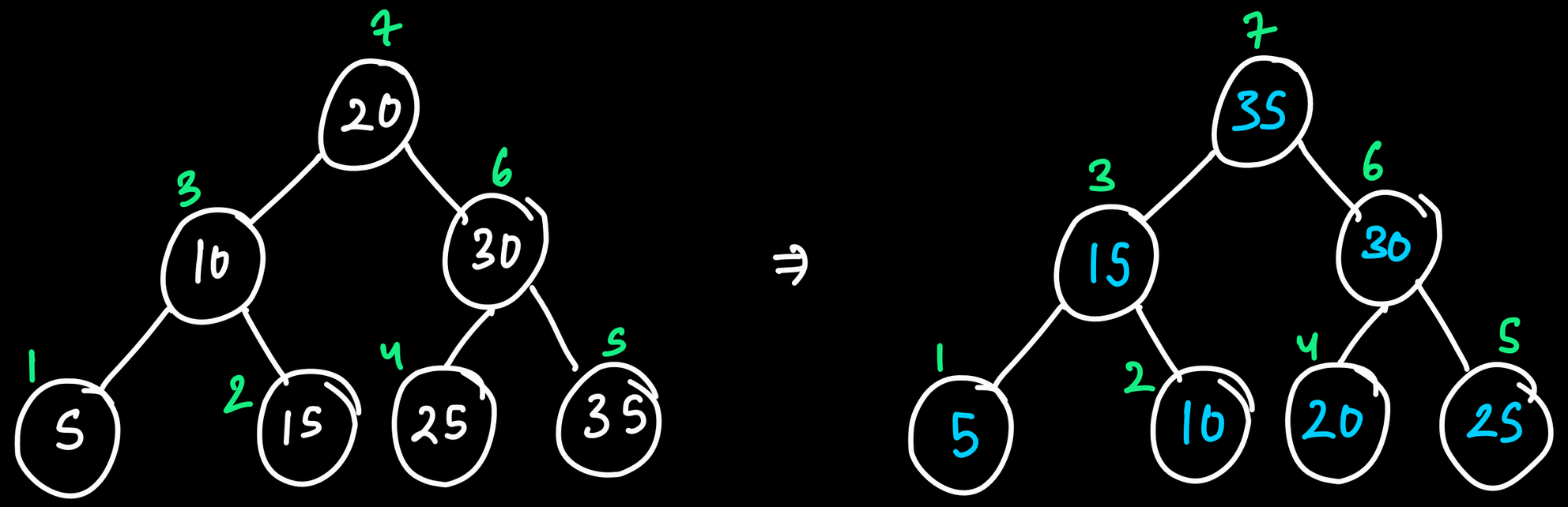
⇒



postorder

1 3 2 5 7 6 4

Ques: BST to Max Heap



in = { 5, 10, 15, 20, 25, 30, 35 }

i

MinHeap Implementation

	0	1	2	3	4	5	6	7
arr	X	9	8	10	15			

i

parent = i/2

↑

root = i

left child = 2i

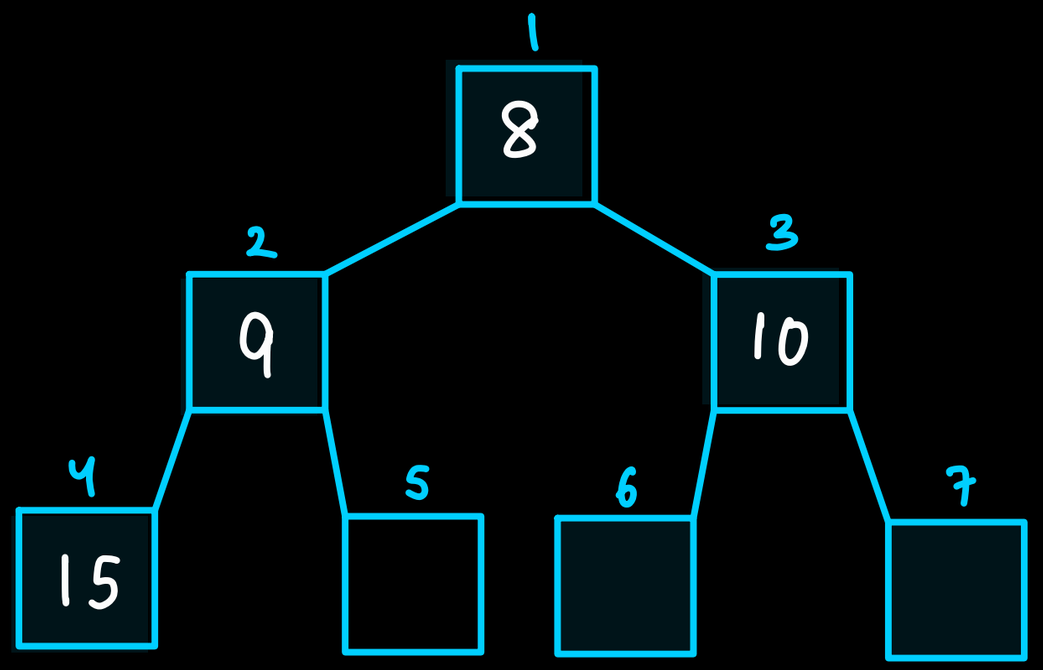
right child = 2i+1

add → rearrange

↓

upar jaayenge

- ✓ add(10)
- ✓ add(15)
- ✓ add(8)
- ✓ add(9)
- ✓ add(4)
- ✓ remove()
- ✓ add(2)
- ✓ remove()



MaxHeap Implementation